MICRO COMPUTER INSTITUTE

Hardware & Networking

Foundation Study Pack - Volume 1

HARDWARE | Duration: 6 Months

Learn - Practise - Apply
A structured bilingual-support handbook for classroom, lab and online learners.

Student use: Read each module, complete the lab activity, save evidence in your portfolio, and discuss doubts with your instructor.

Hindi support (Roman): Har module ko padhen, practical lab activity poori karen, apna work save karen aur doubt teacher se poochhen.

Course Roadmap

This 6 Months programme is organised into 8 learning modules. The sequence may be adjusted by the instructor to match the batch calendar.

No.	Module	Learning focus
1	PC Components	Motherboard, CPU, RAM, storage, power supply and peripherals.
2	Assembly	Compatibility, ESD safety, installation and cable management.
3	Operating Systems	Installation, drivers, updates, recovery and backup.
4	Troubleshooting	Symptoms, diagnostics, replacement strategy and documentation.
5	Networking Basics	LAN, WAN, IP addressing, switches, routers and Wi-Fi.
6	Network Setup	Cabling, sharing, router configuration and basic commands.
7	Security & Maintenance	Updates, antivirus, access control and preventive care.
8	Service Project	Assemble, configure, test and document a small office network.

Module 1: PC Components

Motherboard, CPU, RAM, storage, power supply and peripherals.

Key ideas

• Explain the purpose and essential vocabulary of pc components.

• Demonstrate the standard workflow accurately and safely.

• Choose suitable tools, settings and file formats for the task.

• Check the output, correct errors and save work with a meaningful name.

Lab activity

Create a small working example that demonstrates pc components. Save the source file and a PDF or screenshot as evidence.

Self-check: Can you repeat the task without step-by-step help and explain why each important step is used?

Module 2: Assembly

Compatibility, ESD safety, installation and cable management.

Key ideas

• Explain the purpose and essential vocabulary of assembly.

• Demonstrate the standard workflow accurately and safely.

• Choose suitable tools, settings and file formats for the task.

• Check the output, correct errors and save work with a meaningful name.

Lab activity

Create a small working example that demonstrates assembly. Save the source file and a PDF or screenshot as evidence.

Self-check: Can you repeat the task without step-by-step help and explain why each important step is used?

Module 3: Operating Systems

Installation, drivers, updates, recovery and backup.

Key ideas

• Explain the purpose and essential vocabulary of operating systems.

• Demonstrate the standard workflow accurately and safely.

• Choose suitable tools, settings and file formats for the task.

• Check the output, correct errors and save work with a meaningful name.

Lab activity

Create a small working example that demonstrates operating systems. Save the source file and a PDF or screenshot as evidence.

Self-check: Can you repeat the task without step-by-step help and explain why each important step is used?

Module 4: Troubleshooting

Symptoms, diagnostics, replacement strategy and documentation.

Key ideas

- Explain the purpose and essential vocabulary of troubleshooting.
- Demonstrate the standard workflow accurately and safely.
- Choose suitable tools, settings and file formats for the task.
- Check the output, correct errors and save work with a meaningful name.

Lab activity

Create a small working example that demonstrates troubleshooting. Save the source file and a PDF or screenshot as evidence.

Self-check: Can you repeat the task without step-by-step help and explain why each important step is used?

Module 5: Networking Basics

LAN, WAN, IP addressing, switches, routers and Wi-Fi.

Key ideas

- Explain the purpose and essential vocabulary of networking basics.
- Demonstrate the standard workflow accurately and safely.
- Choose suitable tools, settings and file formats for the task.
- Check the output, correct errors and save work with a meaningful name.

Lab activity

Create a small working example that demonstrates networking basics. Save the source file and a PDF or screenshot as evidence.

Self-check: Can you repeat the task without step-by-step help and explain why each important step is used?

Module 6: Network Setup

Cabling, sharing, router configuration and basic commands.

Key ideas

- Explain the purpose and essential vocabulary of network setup.
- Demonstrate the standard workflow accurately and safely.
- Choose suitable tools, settings and file formats for the task.
- Check the output, correct errors and save work with a meaningful name.

Lab activity

Create a small working example that demonstrates network setup. Save the source file and a PDF or screenshot as evidence.

Self-check: Can you repeat the task without step-by-step help and explain why each important step is used?

Module 7: Security & Maintenance

Updates, antivirus, access control and preventive care.

Key ideas

• Explain the purpose and essential vocabulary of security & maintenance.

• Demonstrate the standard workflow accurately and safely.

• Choose suitable tools, settings and file formats for the task.

• Check the output, correct errors and save work with a meaningful name.

Lab activity

Create a small working example that demonstrates security & maintenance. Save the source file and a PDF or screenshot as evidence.

Self-check: Can you repeat the task without step-by-step help and explain why each important step is used?

Module 8: Service Project

Assemble, configure, test and document a small office network.

Key ideas

• Explain the purpose and essential vocabulary of service project.

• Demonstrate the standard workflow accurately and safely.

• Choose suitable tools, settings and file formats for the task.

• Check the output, correct errors and save work with a meaningful name.

Lab activity

Create a small working example that demonstrates service project. Save the source file and a PDF or screenshot as evidence.

Self-check: Can you repeat the task without step-by-step help and explain why each important step is used?

Practical Workbook

Complete these tasks in order. The instructor may adapt names and sample data, but every learner should produce original work.

Practical 1. Create and save a correctly named practice file for PC Components. Include a short note describing what you learned.

Practical 2. Create and save a correctly named practice file for Assembly. Include a short note describing what you learned.

Practical 3. Create and save a correctly named practice file for Operating Systems. Include a short note describing what you learned.

Practical 4. Create and save a correctly named practice file for Troubleshooting. Include a short note describing what you learned.

Practical 5. Create and save a correctly named practice file for Networking Basics. Include a short note describing what you learned.

Practical 6. Create and save a correctly named practice file for Network Setup. Include a short note describing what you learned.

Portfolio checklist

- Use folder format: CourseCode / StudentName / Module / Activity.
- Keep editable source files plus exported PDF or screenshot.
- Do not copy another student work; mention sources used.
- Back up important work in two locations.

Assignments & Assessment

Assignment 1: Concept Check

Write 300-500 words explaining three important concepts from PC Components and Assembly. Add one real-life example.

Assignment 2: Practical Submission

Complete a combined task using Operating Systems and Troubleshooting. Submit source file, output and a one-page process note.

Assignment 3: Problem Solving

Identify a realistic institute or small-business problem that can be improved using Hardware & Networking. Propose steps, tools, risks and success measures.

Assignment 4: Final Project

Build a complete project using at least four course modules. Test it, prepare screenshots, and present the result in 5-7 minutes.

Criterion	Weight
Accuracy and completeness	30%
Practical application	30%
Presentation and file organisation	20%
Originality, explanation and safe practice	20%

Revision & Safe Learning Guide

- Review class notes within 24 hours and practise again without watching the demonstration.
- Use legal software and trusted downloads. Scan external drives and downloaded files.
- Never share passwords, OTPs, private documents or student data.
- Verify AI-generated or internet information before using it in assignments.
- Use clear filenames, version numbers and regular backups.
- Ask for help when an error could damage data, equipment or accounts.

Quick viva questions

1. What is the purpose of PC Components, and what common mistake should a beginner avoid?
2. What is the purpose of Assembly, and what common mistake should a beginner avoid?
3. What is the purpose of Operating Systems, and what common mistake should a beginner avoid?
4. What is the purpose of Troubleshooting, and what common mistake should a beginner avoid?
5. What is the purpose of Networking Basics, and what common mistake should a beginner avoid?
6. What is the purpose of Network Setup, and what common mistake should a beginner avoid?

Support

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